

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Eighth Semester B.E. (Credit System) Degree Examinations
 May - June 2023

19ME8X28 – OPERATIONS MANAGEMENT AND ENTREPRENEURSHIP

Duration: 3 Hours

Max. Marks: 100

- Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.
 2) Use of SQC Table is permitted.

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) With example explain the classification of production system. | 10 | L*2 | 1 | 1 |
| b) Write brief notes on the following. | | | | |
| i) Kaizen | | | | |
| ii) Juran's Quality Trilogy | 10 | L2 | 2 | 1 |
| 2. a) What are the objectives of production management and operations managements? | 10 | L2 | 1 | 1 |
| b) Define TQM. Explain characteristics / nature of TQM. | 10 | L2 | 2 | 1 |
| 3. a) Explain the framework of operations management. | 10 | L2 | 2 | 1 |
| b) Explain the 7 basic tools used in quality control. | 10 | L2 | 2 | 1 |

Unit – II

4. a) The following are the values of the mean ($\bar{X}$) and the range (R) for ten samples of size 5 each. Draw the mean chart and comment on the state of control of the process.

Sample No	1	2	3	4	5	6	7	8	9	10
$\bar{X}$	43	49	37	44	45	37	51	46	43	47
R	5	6	5	7	7	4	8	6	4	6

Given the following control chart constraint for $n=5$, $A_2=0.58$, $D_3=0$ and $D_4=2.115$.

- | | | | | |
|---|--------|----|---|---|
| b) What are intrapreneurs? List the differences between intrapreneurs & Entrapreneurs. | 10 | L3 | 3 | 2 |
| 5. a) Explain Type I and Type II error with example. | 10 | L2 | 3 | 1 |
| b) What are the four stages in entrepreneurial process? | 4 | L1 | 4 | 1 |
| c) What are the roles that entrepreneurship plays in the economical development of a country? | 6 | L1 | 4 | 1 |
| 6. a) Discuss the concept of Six Sigma with an example. | 10 | L2 | 3 | 1 |
| b) What are the functions of an entrepreneur? Explain different types of entrepreneurs. | 10 | L1 | 3 | 1 |
| Unit – III | | | | |
| 7. a) List the objectives and functions of KIADB. | 6 | L2 | 5 | 1 |
| b) Define small scale industry. Write the characteristics of small-scale industries. | 7 | L2 | 5 | 1 |
| c) Explain the steps required for establishing an SSI. | 7 | L2 | 5 | 1 |
| 8. Write short note on | | | | |
| i) SISI ii) SIDBI iii) KSFC iv) NSIC | 5x4=20 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
